Norms

Body text on white paper. Each inner product determines a norm; working with norms squared is usually easier than working directly with the norms themselves.

6.7 definition: norm

For v in V, the norm of v, denoted $\|v\|$, is defined by

$$\|v\| = \sqrt{\langle v, v\rangle}.$$

6.9 basic properties of the norm

Suppose v in V.

(a) $\|v\| = 0$ if and only if $v = 0$.

(b) $\|\lambda v\| = |\lambda|\,\|v\|$ for all λ in F.

6.10 definition: orthogonal

Two vectors u, v in V are called orthogonal if $\langle u, v\rangle = 0$.

6.11 orthogonality and 0

(a) 0 is orthogonal to every vector in V.

(b) 0 is the only vector in V orthogonal to itself.

6.12 Pythagorean theorem (green)

Suppose u, v in V. If u and v are orthogonal, then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$.

Cross-references: see Thm 2.5 and the colored ref §4.1.

Color swatches (how the filter maps pure colors):

white	black	red	green	blue	cyan	magenta	yellow	paper	ink

Color figure (phase portrait) — how images survive the filter

A domain-coloring image like the phase portraits in Visual Complex Functions. Under a dark (invert) theme, photographic color inverts to its complement — worth seeing.

Table with a colored header row:

n	norm	squared
1	1.00	1.00
2	1.41	2.00
3	1.73	3.00